

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

Nov-Dec 2012

EE416 Advanced Microcontrollers & Applications

Date: 08.12.2012, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. Write comments for each line of C program.

SECTION - I

- Q - 1 (a) List out all the peripherals available in 8051 microcontroller. [2]
- (b) Draw hardware for key interfacing & explain in brief. [3]
- (c) Explain the necessary steps to calculate the number to be loaded in TLx & THx registers for given delay & crystal frequency for 16 bit timer? [2]
- Q - 2 (a) List out the special on chip features available in Advanced microcontroller. [4]
- (b) For microcontroller 8051, list out various modules of PCA timer & calculate number to be loaded in PCA timer register for overflow rate of 10 KHz. [2]
- (c) Compare C language programming & Assembly language programming of microcontroller. [4]
- (d) Explain various control loops used in C Programming of microcontroller with example program. [4]

OR

- Q - 2 (a) Explain various operators used in C programming of microcontroller assuming required variables. Show both byte wise & bit wise operation. [4]
- (b) Explain general structure of a C language microcontroller program. [4]
- (c) Explain key words sbit, sfr, sfr16 & interrupt used in C programming on Keil compiler. [2]
- (d) Derive the necessary code for displaying 0 to 9 for both common anode & common cathode configuration of seven segment LED display. [4]
- Q - 3 (a) Compare Port pin structure of 8051 & c8051f120 microcontroller. [2]
- (b) Explain the concept of cross bar decoder & SFR paging in CIP-51 microcontroller. [4]

- (c) Explain 16 bit PWM mode of operation (Using PCA timer module) for c8051f120 microcontroller. Write down program with comments for generating 10 KHz, 60 % duty cycle waveform. Show necessary calculations. [8]

OR

- Q - 3 (a) List out various mode of operation possible for timer 2 of c8051f120. Explain toggle mode of operation. [6]
- (b) Explain role of watchdog timer. Write necessary code to disable it. [2]
- (c) Write a program with comments for c8051f120 microcontroller to generate 20 %, 20 KHz signal on port pin P1.0 using timer 0. Show necessary calculations. [6]

SECTION – II

- Q - 4 (a) Write a program which measures a dc signal available on AIN 0.1 pin & makes port pin LOW if this voltage exceeds 2 Volt level. The Pin status should not be changed once the 2 Volt limit is crossed. [5]
- (b) Write down differences between ADC0 & ADC2 module of c8051f120 microcontroller. [2]
- Q - 5 (a) The DC motor speed control scheme is required to be designed with c8051f120 microcontroller. Draw complete hardware design. List out the SFR register required to be used in Program with brief reason. [6]
- (b) Write a program to generate triangular voltage of 1000 Hz, using DAC0. Show necessary calculations & state your assumption. [4]
- (c) If reference voltage to ADC0 is 3 Volt, Find out the content of ADC0H & ADC0L for the application of 0.5 Volt, 1 Volt, 2.62 Volt & 1.59 Volt. Find out for both right justification & left justification assuming PGA gain to be unity. [4]

OR

- Q - 5 (a) Write a portion of program to update DAC output on overflow of timer 0. The DAC output is available by writing in to DAC0H & DAC0L registers. [4]
- (b) Write a program to generate staircase voltage of 1000 Hz, using DAC0. Show necessary calculations & state your assumption. [4]
- (c) A simple single phase over current relay is required to be designed on c8051f120 microcontroller. Draw suitable hardware. List out the SFR registers required to be used in program with reason in brief. [6]
- Q - 6 (a) Explain Ready, Running, Deleted, Waiting & Time out state of Task. [5]
- (b) Explain the difference between followings. [2]

1. Multi tasking & Multiprocessing.
2. Hard real time & Soft real time OS.

- (c) Explain the execution of following RTX51 based program written on Keil compiler. [7]

```
#include<rtx51tny.h>

int counter0, counter1;

void job0( void ) _task_0 {
    os_creat(1);
    while(1) {
        if(++counter0 == 0 )
            os_send_signal(1);} }

void job1( void ) _task_1 {
    while(1) {
        os_wait (K_SIG, 0, 0);
        counter1++; } }
```

OR

- Q - 6 (a) Write & explain a simple program to explain task switching in round robin architecture. [2]
- (b) Explain following in built function of RTX51 RTOS. [12]

1. lsr_send_signal()
2. Os_clear_signal()
3. Os_creat_task()
4. Os_delete_task()
5. Os_send_signal()
6. Os_wait()

Write a sample short program to explain each function.
